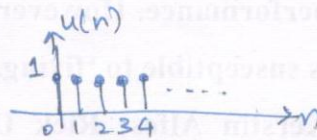


Special sequences

I. Unit-step sequence:

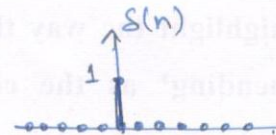
$$u_n = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$$



II unit Impulse sequence:

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

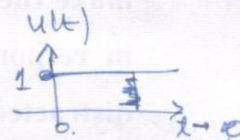
$$= u_n - u_{n-1}$$



III unit-Ramp sequence: $r(n) = n \cdot u(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$

Q. ~~x(t) = u(t)~~ for a cts unit-step fn

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & 0 \leq t < \infty \end{cases}$$



and $u(-t) = \begin{cases} 1, & 0 < -t < \infty \Rightarrow -\infty < t < 0 \\ 0, & -t < 0 \Rightarrow t > 0 \end{cases}$

For a function with both causal & anti-causal part.

$$f(t) = f(t)u(t) + f(t)u(-t)$$

Whereas for a discrete time signal/function:

$$f[n] = f[n]u[n] + f[n]u[-n-1]$$

Proof: taking $n \rightarrow -(n+1)$

$$u[-(n+1)] = \begin{cases} 1, & -(n+1) \geq 0 \Rightarrow -n \geq 1 \Rightarrow n \leq -1 \\ 0, & \text{otherwise} \end{cases}$$

Frequency Transformations:

- Goal of DSP is usually to measure, filter, compress continuous real-world analog signals.
- So first step is to convert signal from analog to digital form → sampling, quantization, encoding.
- Then study these digital signals in different domains.
 - time domain
 - spatial " (multi-dimensional signals)
 - frequency "
 - wavelet. "
- Freq. domain analysis - used in conditions where processes such as filtering, amplifying & mixing are required.
- Freq. domain - refers to analysis of mathematical functions / signals w.r.t. frequency.

Freq. domain graphs shows how much of the signal lies ~~not~~ within given frequency band over a range of frequencies.

→ For continuous signals:

- ① Fourier Series - decomposes periodic functions/signals into a sum of a set of simple oscillating functions i.e. sin & cos. P-Period

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^N \left[a_n \cos\left(\frac{2\pi n x}{P}\right) + b_n \sin\left(\frac{2\pi n x}{P}\right) \right]$$

- ② Fourier Transform - used for Periodic / aperiodic signals

$$F(\omega) = \int_{-\infty}^{\infty} f(x) \cdot e^{-j2\pi\omega x} dx$$

$$f(x) = \int_{-\infty}^{\infty} F(\omega) e^{j2\pi\omega x} d\omega$$

Reversible function

③ Laplace Transform - used for solving differential & integral equations by converting them into linear equations.

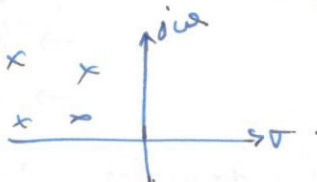
- used for analysis of LTI systems such as electrical circuits, harmonic oscillations, optical devices & mechanical systems.

- It tells about the stability of the system.

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt.$$

defined of all real nos. $t \geq 0$.

s - complex no. $= \sigma + j\omega$



For Discrete-Time Signals.

④

DTFT - Discrete Time FT.

Resultant DTFT frequency-domain representation is always a periodic function. $-\pi \leq \omega \leq \pi$.
Period of 2π .

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

↑
continuous-time FT.

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) \cdot e^{j\omega n} d\omega.$$

⑤ DFT - Sampling of DTFT with $\omega = \frac{2\pi}{N} \cdot k$.

- It deals with finite amt of data.

N : width of 1 sample.
 k : no. of sample.

$$X\left(\frac{2\pi}{N} \cdot k\right) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N} k \cdot n}.$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{-j\frac{2\pi}{N} k \cdot n}.$$

⑥ Z-transform - like Laplace, used for discrete-time signals.

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

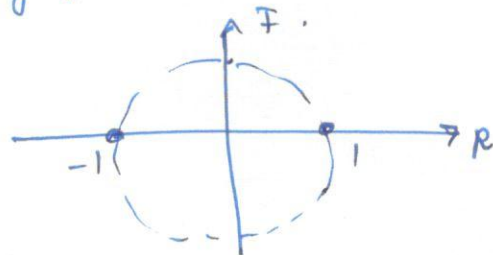
$$z = A e^{j\phi} = A (\cos \phi + j \sin \phi)$$

A - magnitude

j - imaginary unit.

ϕ - complex argument (phase / angle).

$$x[n] = \mathcal{Z}^{-1}\{X(z)\} = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$



Z - TRANSFORM.

(1)

- used in the analysis of and characterization of linear time - Invariant systems.
- Applied on discrete - time signals.

$$X(z) \triangleq \sum_{n=-\infty}^{\infty} x[n] z^n$$

$x[n]$: discrete-time signal.

i.e. infinite series.

- but it exists only for those values of z , for which the series converges.

- i.e. Region of Convergence (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value.

Ex.

$$x_1(n) = [1, 2, 5, 7, 0, 1]$$

$$\text{Then } X_1(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$

ROC: entire z -plane, except $z=0$.

$$x_2(n) = [1, 2, 5, 7, 0, 1]$$

$$X_2(z) = z^2 + 2z^1 + 5 + 7z^{-1} + z^{-3}$$

ROC: entire z -plane, except $z=0$ & $z=\infty$.

Ex.

$$x_3(n) = \delta(n)$$

$$X(z) = 1$$

ROC: entire z -plane.

$$x(n) = \delta(n-k), k > 0$$

$$X(z) = z^{-k}$$

ROC: entire z -plane except $z=0$.

$$x(n) = \delta(n-k), k \leq 0$$

Ex.

$$x(n) = \left(\frac{1}{2} \right)^n \cdot u(n)$$

$$x(n) = \left[1, \left(\frac{1}{2} \right), \left(\frac{1}{2} \right)^2, \left(\frac{1}{2} \right)^3, \dots, \left(\frac{1}{2} \right)^n, \dots \right]$$

$$X(z) = 1 + \frac{1}{2}z^{-1} + \left(\frac{1}{2} \right)^2 z^{-2} + \left(\frac{1}{2} \right)^3 z^{-3} + \dots$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2} \right)^n \cdot z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2} \cdot z^{-1} \right)^n$$

GP.

$$1 + A + A^2 + A^3 + \dots = \frac{1}{1-A}, \quad |A| < 1.$$

$$\text{Therefore, } \left| \frac{1}{2} z^{-1} \right| < 1 \Rightarrow |z^{-1}| < \frac{1}{2}$$

$$\Rightarrow |z| > \frac{1}{2}$$

Also.

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad \text{ROC: } |z| > \frac{1}{2}$$

~~Second Alternate Representation~~

Now, $z = re^{j\theta}$ polar form.

where $r = |z|$, $\theta = \angle z$.

$$\therefore X(z) \Big|_{z=re^{j\theta}} = \sum_{n=-\infty}^{\infty} x(n) \cdot r^{-n} \cdot e^{-j\theta n}$$

Now, ROC defines $|X(z)| < \infty$.

$$\begin{aligned} \therefore |X(z)| &= \left| \sum_{n=-\infty}^{\infty} x(n) r^{-n} \cdot e^{-j\theta n} \right| \leq \sum_{n=-\infty}^{\infty} |x(n) r^{-n} e^{-j\theta n}| \\ &= \sum_{n=-\infty}^{\infty} |x(n) r^{-n}| \end{aligned}$$

Now for $|x(z)| < \infty$, $x(n)r^{-n}$ sequence should be absolutely summable. $\rightarrow$ ~~that~~ found out from

$\rightarrow$ range of values of r for which its summable.

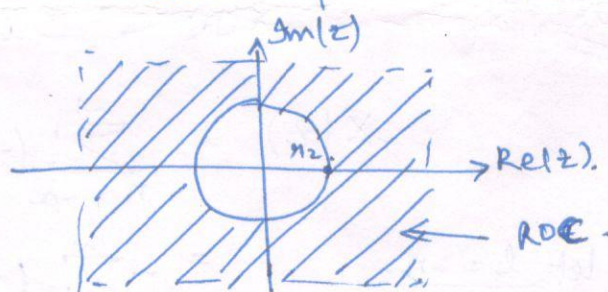
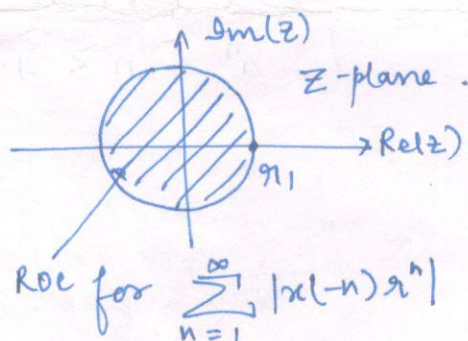
$$|x(z)| \leq \sum_{n=-\infty}^{-1} |x(n)r^{-n}| + \sum_{n=0}^{\infty} \left| \frac{x(n)}{r^n} \right|$$

$$\leq \underbrace{\sum_{n=1}^{\infty} |x(-n)r^n|}_{\text{for } 1 \leq n \leq \infty} + \underbrace{\sum_{n=0}^{\infty} \left| \frac{x(n)}{r^n} \right|}_{\text{for this series to be convergent}}$$

$$\begin{aligned} n &= -n \\ \sum_{n=-\infty}^{-1} x(n)r^{-n} &= \sum_{n=1}^{\infty} x(-n)r^n \\ \sum_{n=1}^{\infty} |x(-n)r^n| \end{aligned}$$

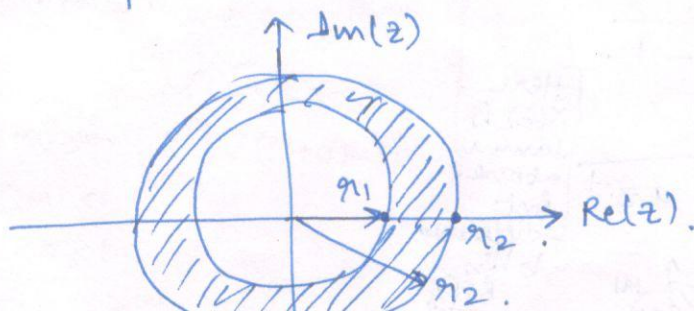
There must exist small values of r to make the series summable. $\therefore$ ROC be all points in a circle of some radius r_1 , $r_1 < \infty$

for this ~~series~~ $x(z)$ to be convergent, this series must have large value of r . for $0 \leq n \leq \infty$. Let r_2 . $\therefore$ ROC - all points outside circle of radius r_2 .



$\rightarrow$ Since $x(z)$ requires convergence of both ~~series~~ summations. $\therefore$ ROC of $x(z)$ is angular region in z -plane $r_2 < r < r_1$. Common region.

$\rightarrow$ If $r_2 > r_1$, there is no common region $\therefore x(z)$ does not exist.



Inverse Z-Transform - : (*) leave.

Transforming z-domain to time domain.

$$X(z) = \sum_{k=-\infty}^{\infty} x(k) z^{-k}$$

suppose we multiply both sides with z^{n-1} & integrate both sides over a closed ~~conv~~ contour within ROC of $X(z)$.

$$\oint_c X(z) z^{n-1} dz = \oint_c \sum_{k=-\infty}^{\infty} x(k) \cdot z^{n-1-k} dz$$

$$\Rightarrow \oint_c X(z) z^{n-1} dz = \sum_{k=-\infty}^{\infty} x(k) \oint_c z^{n-1-k} dz$$

c: closed contour in ROC of $X(z)$ taken in counterclockwise direction.

Cauchy integral theorem.

$$\text{i.e. } \frac{1}{2\pi j} \oint_c z^{n-1-k} dz = \begin{cases} 1 & , k=n \\ 0 & , k \neq n \end{cases}$$

$$\Rightarrow x(n) = \frac{1}{2\pi j} \oint_c X(z) \cdot z^{n-1} dz$$

Inverse Z-transform:

$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

Integral - contour integral
over a closed path

C , enclosing origin
& lies within ROC of $X(z)$

3 Methods to solve:

i. Direct evaluation
by contour integration.

⇒ Cauchy Residue theorem. Do yourself.

ii. Direct Long Division Method:

eg: $X(z) = \frac{z^2 + z}{z^2 - 3z + 4}$

Represent $X(z)$ in terms of z^{-1} .

$$X(z) = \frac{1 + z^{-1}}{1 - 3z^{-1} + 4z^{-2}}$$

$$\begin{array}{r} 1 - 3z^{-1} + 4z^{-2} \overline{) 1 + z^{-1}} \\ \underline{1 + z^{-1}} \phantom{+ 4z^{-2}} \\ 4z^{-1} - 4z^{-2} \\ \underline{4z^{-1} - 12z^{-2} + 16z^{-3}} \\ 8z^{-2} - 16z^{-3} \\ \underline{8z^{-2} - 24z^{-3} + 32z^{-4}} \\ 8z^{-3} - 32z^{-4} \\ \underline{8z^{-3} - 32z^{-4}} \\ 0 \end{array}$$

$$\therefore x(0) = 1, \quad x(1) = 4, \quad x(2) = 8, \quad x(3) = 8, \dots$$

iii.

Partial fraction Expansion.Ex.

$$H(z) = \frac{-4+8z^{-1}}{1+6z^{-1}+8z^{-2}}.$$

$$= \frac{-4+8z^{-1}}{(1+4z^{-1})(1+2z^{-1})}$$

$$= \frac{A_1}{1+4z^{-1}} + \frac{A_2}{1+2z^{-1}}$$

$$A_1 = \left. \frac{-4+8z^{-1}}{1+2z^{-1}} \right|_{\text{at } z^{-1} = -1/4} = \frac{-6}{1/2} = -12.$$

$$A_2 = \left. \frac{-4+8z^{-1}}{1+4z^{-1}} \right|_{\text{at } z^{-1} = -1/2} = \frac{-8}{-1} = 8.$$

$$\therefore H(z) = \frac{-12}{1+4z^{-1}} + \frac{8}{1+2z^{-1}}$$

Taking inverse z-transform.

$$h(n) = [-12(-4)^n + 8(-2)^n] u(n).$$

Poles & zeros.

In General, $X(z)$ takes the form of rational function of z .

$$\begin{aligned} \text{i.e. } X(z) &= \frac{b_0 z^m + b_1 z^{m-1} + \dots + b_m}{z^n + a_1 z^{n-1} + \dots + a_n} \\ &= \frac{b_0 (z - z_1)(z - z_2) \dots (z - z_m)}{(z - p_1)(z - p_2) \dots (z - p_n)}. \end{aligned}$$

p_i - poles of $X(z)$, z_i - zeros of $X(z)$

In z -domain $X(z)$ is denoted as a function of z^{-1} .

$$X(z^{-1}) = \frac{b_0 z^{-(n-m)} + b_1 z^{-(n-m+1)} + \dots + b_m z^{-n}}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_n z^{-n}}.$$

For example.

$$X(z) = \frac{z^2 + 2z}{z^2 + 3z + 2} = \frac{z(z+2)}{(z+1)(z+2)}$$

$\Rightarrow$ Poles : $z = -1, z = -2$.
Zeros : $z = 0, z = -2$

$$X(z^{-1}) = \frac{1 + 2z^{-1}}{1 + 3z^{-1} + 2z^{-2}} = \frac{(1 + 2z^{-1})}{(1 + z^{-1})(1 + 2z^{-1})}$$

Poles : $z = -1, z = -2$.

Zeros : $z = 2$.

Not expressing $z = 0$.

So it's better to express $X(z)$ as fn of z .

System function of a linear time-invariant system.

$$y(n) = h(n) * x(n)$$

$$\Rightarrow Y(z) = H(z) X(z)$$

$h(n)$ - Impulse Response

$H(z)$ - Transfer fn of system.

If we know $x(n)$ & observe o/p, $y(n)$ of system.

Then Transfer fn of system

$$H(z) = \frac{Y(z)}{X(z)}$$

$$H(z) = \sum_{n=-\infty}^{\infty} h(n) \cdot z^{-n}$$

Let, the system is described by a linear constt-coefficient difference equation of form,

$$y(n) = + \sum_{k=1}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

$$\Rightarrow Y(z) = + \sum_{k=1}^N a_k Y(z) \cdot z^{-k} = \sum_{k=0}^M b_k X(z) \cdot z^{-k}$$

$$\Rightarrow Y(z) = \left[1 + \sum_{k=1}^N a_k z^{-k} \right] = X(z) \left[\sum_{k=0}^M b_k z^{-k} \right]$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = H(z) \quad \therefore \text{Pole-zero system.}$$

2 cases : ① If $a_k = 0, 1 \leq k \leq N$.

Then $H(z) = \sum_{k=0}^M b_k z^{-k}$

: All zero system.
finite duration
impulse response
(FIR)

is also called

(2)

if $0k=0$, $1 \leq k \leq M$.

: All pole system

$$H(z) = \frac{b_0}{1 + \sum_{k=1}^N a_k z^{-k}}$$

IR

Ans. Consider diff. eqn.

$$y(k+2) + a_1 y(k+1) + a_2 y(k) = b_0 u(k+2) + b_1 u(k+1) + b_2 u(k)$$

$$u(k) = 0, k < 0$$

find Pulse Transfer Fn

~~$$z^2 Y(z) + z a_1 Y(z) + a_2 Y(z) =$$~~

Property:

$$\text{If } x(k) = 0, k < 0$$

Then

$$\mathcal{Z}\{x(k+d)\} = z^d X(z) - \sum_{i=1}^d x(d-i) z^i$$

$$= z^d X(z) - z^d x(0) - z^{d-1} x(1) \dots - z x(d-1)$$

$$\Rightarrow [z^2 Y(z) - z^2 y(0) - z y(1)] + a_1 [z Y(z) - z y(0)] + a_2 Y(z) = b_0 [z^2 U(z) - z^2 u(0) - z u(1)] + b_1 [z U(z) - z u(0)] + b_2 U(z)$$

$$\Rightarrow [z^2 + a_1 z + a_2] Y(z) = (b_0 z^2 + b_1 z + b_2) U(z) + z^2 [y(0) - b_0 u(0)] + z [y(1) + a_1 y(0) - b_0 u(1) + b_1 u(0)]$$

$$\Rightarrow Y(z) = \frac{b_0 z^2 + b_1 z + b_2}{z^2 + a_1 z + a_2} U(z) + \frac{[z^2 [y(0) - b_0 u(0)] + z [y(1) + a_1 y(0) - b_0 u(1) + b_1 u(0)]]}{[]}$$

Putting $k = -2$, in original eq. to get initial conditions

$$\Rightarrow y(0) = b_0 u(0)$$

$$k = -1 \Rightarrow y(1) = -a_1 y(0) + b_0 u(1) + b_1 u(0)$$

$$Y(z) = \frac{b_0 z^2 + b_1 z + b_2}{z^2 + a_1 z + a_2} U(z) \Rightarrow G(z) = \frac{b_0 z^2 + b_1 z + b_2}{z^2 + a_1 z + a_2}$$

SYSTEMS.

An arrangement or inter connection of elements which intakes energy, processes it & outputs energy in the desired form.

cts-time

↳ associated signals are continuous

Discrete time.

↳ signals are discrete

Further classification.

1. Static & Dynamic System

Static - o/p of this system at any specific time depends on i/p at that particular time & not on past or future values of i/p.
i.e. - system with no memory.

e.g. resistive network.
(~~Robust~~ system) ↳ i/o relation of such system does not involve integrals or derivatives.

Dynamic - o/p - depends upon i/p at that specific time & at other times.

e.g. any electric ckt consisting of a capacitor or an inductor.

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + y(t) = \frac{d x(t)}{dt} + 2 x(t).$$

② Linear & Non-Linear:

LTI — electrical ckt
consisting of resistances,
C & Inductors

Linear — is one which obeys the principle of superposition (i.e. homogeneous & additive).
 $a y_1(t) + b y_2(t) = a x_1(t) + b x_2(t)$
 Scaling.

Q.g. $y(n) = x^2(n)$

if. $x(n) = 0$, $y(n) = 0$ — system is homogeneous.

$$y(n) = [x_1(n) + x_2(n)]^2$$

$$= x_1^2(n) + x_2^2(n) + 2x_1(n)x_2(n)$$

$$= y_1^2(n) + y_2^2(n) + \dots \neq y_1^2(n) + y_2^2(n)$$

Not additive.

Ans.

$$\frac{dy(t)}{dt} + 2y(t) = x(t)$$

→ Linear / Non

Let for an I/P $x_1(t)$ — response $y_1(t)$
 $x_2(t)$ — $y_2(t)$.

$$\frac{dy_1(t)}{dt} + 2y_1(t) = x_1(t)$$

$$\frac{dy_2(t)}{dt} + 2y_2(t) = x_2(t)$$

$$a \frac{dy_1(t)}{dt} + b \frac{dy_2(t)}{dt} + 2ay_1(t) + 2by_2(t) = ax_1(t) + bx_2(t)$$

$$\Rightarrow \frac{d}{dt} (ay_1(t) + by_2(t)) + 2(ay_1(t) + by_2(t)) = ax_1(t) + bx_2(t)$$

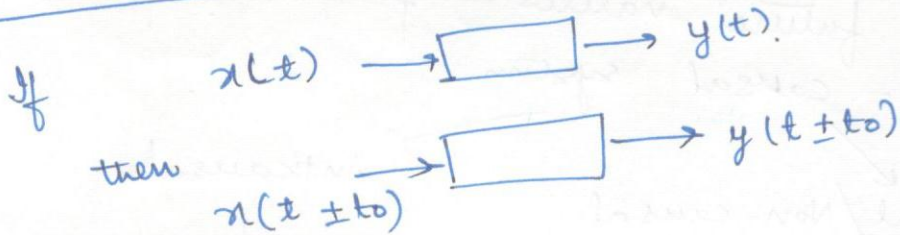
— linear as superposition theorem is satisfied.

Ans

$$\frac{dy(t)}{dt} + 2y(t) + 4 = x(t)$$

— Non-Linear

③ Time-variant & Invariant



ep. — TIV — clicking a picture

TV → human vocal tract
 Transfer fn ~~change~~ any time is depends on shape of vocal organs

If time shifted I/P is applied to a system which produces a time shift of same amt as in I/P, then system is time invariant.

ex. $y(t) = x(2t)$

$y(t-t_0) = x(2t-2t_0)$ — Delay is different.

If system is linear & Time-invariant, then it is called Linear time-Inv. (LTI) system.

④ Causal & Non-Causal syst.

↓
 Response of causal system to an input does not depend on future values of that I/P, but depends only on the present & / or past values.
 — also known as physical / non-anticipitative system.

— ~~impulse~~ Response of system must be 0 $\forall t < 0$

leave $y(t) = \int_{-\infty}^{\infty} \sin(t+t_0) \leftarrow \text{Non-causal}$

— e.g. all system mobile phones — ring for only current call

Non-Causal - A system whose present response depends on future values of the I/P is called non-causal system.

↓
acausal / Non-causal
↓
depends on future / past / present

Anticausal.
↓
only on future.

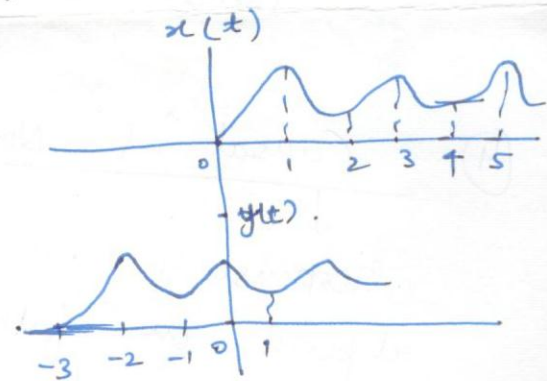
- ~~No~~ No physical system is non-causal in nature

- But can be made realisable in virtual environments.

- Mathematically. if $y(t) = x(t+3)$.
for $t=1$
 $y(1) = x(4)$.

- e.g. recorded cricket match.

- If somehow, depending on past behaviours of any system, we start predicting future values of data & system starts working on it.



⑤ Stable & unstable systems:
 ↳ BIBO - every bounded I/P (finite amplitude) produces a bounded O/P (O/P that doesn't grow unreasonably large).
 (Robust system).
 - Pendulum.

unstable
(fragile)

Nuclear reactions
can become unstable
(uncontrolled)